

Title of Project: Real-Time Imaging for Covid-19 & Breast Cancer Detection

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Abstract: Medical imaging plays a crucial role in disease diagnosis by enabling non-invasive analysis of internal structures. This paper presents a deep learning-based framework for automated classification of COVID-19 using chest CT images and breast lesions using ultrasound imaging. EfficientNetB0 with transfer learning is employed for COVID-19 detection, while a lightweight Convolutional Neural Network (CNN) is used for breast cancer classification into benign, malignant, and normal categories.

The system follows an end-to-end pipeline including preprocessing, training, inference, and deployment through a Flask-based web application that provides real-time predictions with confidence scores. Edge detection techniques are incorporated for basic interpretability, and a SQLite database is used for logging results.

To further improve model robustness and reduce overfitting, data augmentation techniques including rotation, flipping, and scaling are used. To guarantee dependability and consistency across datasets, the models are assessed using important performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix. To boost performance, hyperparameter tuning is used to tune parameters like learning rate, batch size, and epochs.

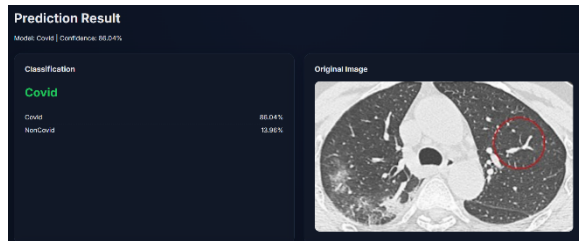
Healthcare practitioners can input medical photos and get immediate diagnostic predictions because to the system's user-friendly interface. Faster processing and adaptability for deployment in contexts with limited resources are guaranteed by its lightweight architecture.

Keywords

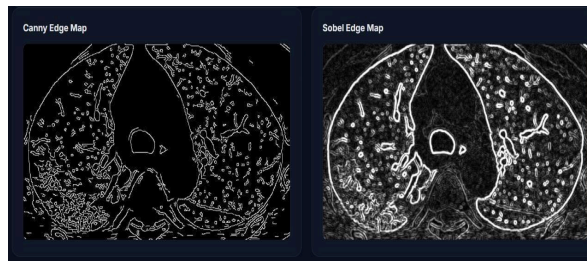
- Deep Learning in Medical Imaging
- COVID-19 and Breast Cancer Classification
- Real-Time Diagnostic System

Photos:

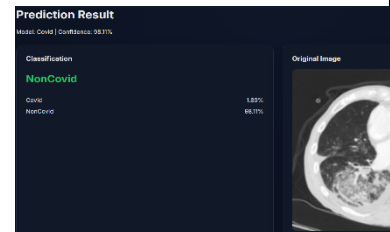
Original image of covid



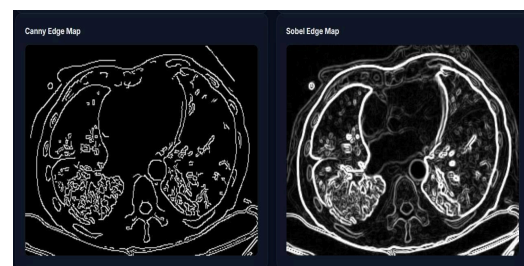
Canny edge map, sobel edge map of covid



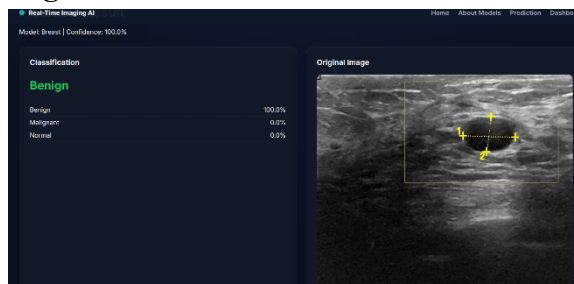
Original image of non-covid



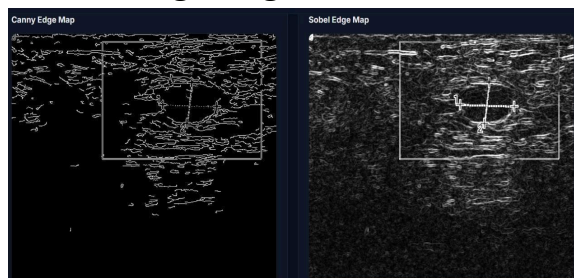
Canny edge map, sobel edge map of non-covid



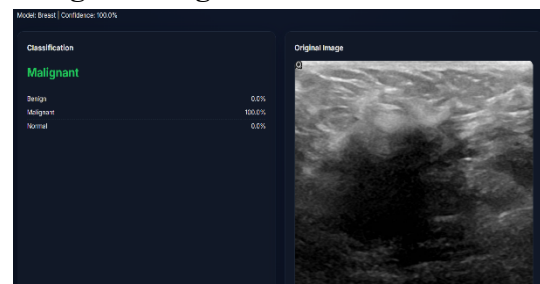
Original image of Breast cancer at Begin stage



Canny edge map, sobel edge map of Breast cancer at Begin stage



Original image of Breast cancer at malignant stage



Canny edge map, sobel edge map of Breast cancer at malignant stage

